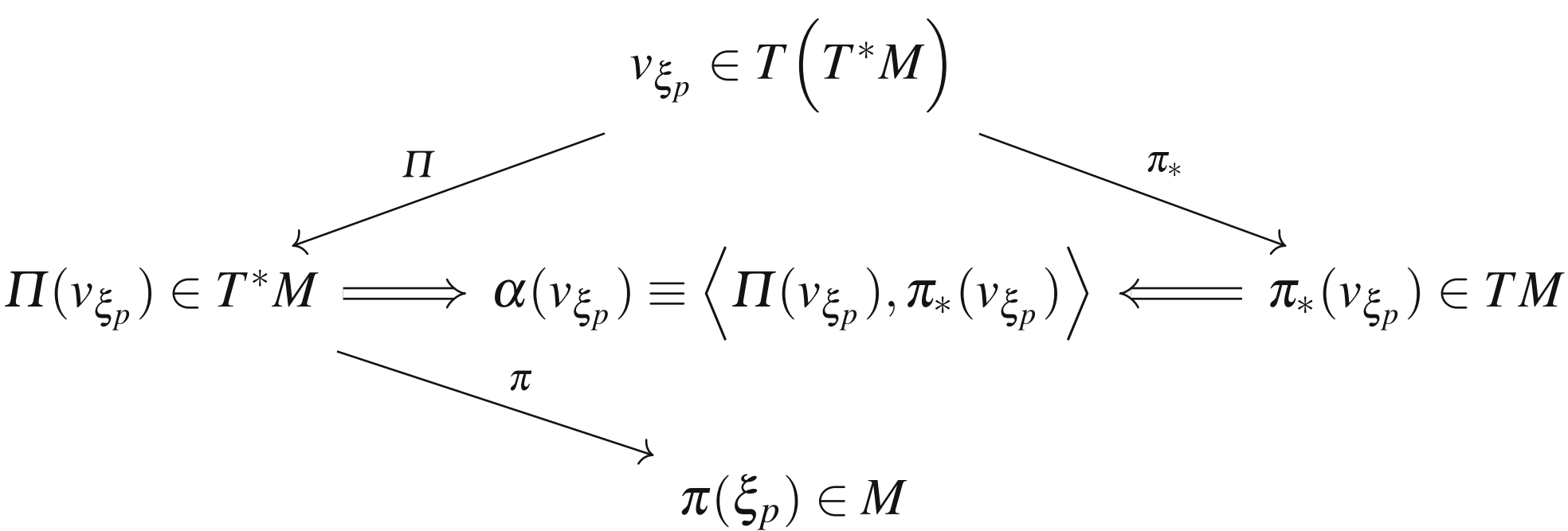

$v_{\xi_p} \in T\left(T^*M\right)$
Π
π_*
$\Pi(v_{\xi_p}) \in T^*M \Longrightarrow \alpha(v_{\xi_p}) \equiv \left\langle \Pi(v_{\xi_p}), \pi_*(v_{\xi_p}) \right\rangle \Longleftarrow \pi_*(v_{\xi_p}) \in TM$
π
$\pi(\xi_p) \in M$